COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010-Data Structures and Algorithms	**SECTION:**	CPE21S2
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Assignment 6.1 Linear and Binary Search

Answer the following questions:

1. What is a search tree in data structures?

- It is a tree based data structure where each node stores a key that is organized in a specific order to make inserting, deleting, and searching data easier and faster compared to an unstructured data.

2. What are the different types of search algorithms in data structures? Differentiate each type of search.

- **Linear/Sequential search**: checks every element one by one from start to finish until the target is found
- **Interval/Binary search**: The data needs to be sorted first because it cuts the search area and half and starts from their for example the data your looking for is the number 8 and you've stored it inside an array that has the values of 1 - 12 it will cut that in half at around 6 and since 8 is greater than 6 it will ignore the lower half and look into the succeeding numbers until it finds the 8.
- **Tree/Graph search**: Just like a family tree it follows a parent to child connection to navigate the data path by using the connection for each node to find the target data.
- **Hash based search**: Uses the hash function which takes a piece of data that it can turn into an address then it can then use to jump directly to the exact location of the data

3. What operations / implementations can be performed using binary and linear search operations?

- **Linear/Sequential search**: Can be used to locate or check a specific element in an unsorted list or array
- **Interval/Binary search**: Can be used to quickly find and locate a specific element in a sorted/arranged array or list.
- **Tree/Graph search:** Can be used to navigate connected data such as finding routes in file directories.
- **Hash based search:** Can be used to quickly locate data using a specific input such as account numbers or product ID's

4. What are the advantages in using binary search trees as data structure?

Since they are structured in an organized order their advantages include:

- Faster searching
- Efficient insertion and deletion
- Keeps data organized
- Easy to traverse

5. Give an example program using binary search and Linear search.

Example of programs that use binary search and linear search include;

Binary search

- **ATM -** finding a customers account number through a sorted list of account numbers

Linear search

- **Game inventory** - Searching through a characters inventory one at a time to find a specific item

CODE

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